# ECEN 689 Homework: Stress-Testing VLMs on 3D Spatial Intelligence

**Course: ECEN 689 Special Topics in Multimodal Computer Vision (Spring 2026), Texas A&M University**
**Submission Format: 1-page summary PDF.** ***Code does not need to be submitted.***

## 1. Context and Goal

**Modern vision-language models (VLMs) describe images and videos effectively but frequently fail on tasks requiring geometry-aware spatial reasoning and navigation-like planning (e.g., estimating metric distance, tracking egocentric directions, and multi-step route planning). Recent evaluations show many video-capable multimodal models fail to beat random chance baselines on spatial intelligence tasks.**

**Your Objective: Identify one or more spatial tasks where VLMs perform poorly, reproduce that weakness through a controlled evaluation, and write a structured analysis of the failure modes. You will compare at least two VLMs and provide evidence-based explanations for their shortcomings.**

## 2. Benchmark and Dataset Background

**This homework builds on the benchmark ecosystem from the paper** ***VLM-3R: Vision-Language Models Augmented with Instruction-Aligned 3D Reconstruction*****. You do not need to reproduce VLM-3R's training; rather, use its evaluation framework and related datasets to stress-test existing models. You must explicitly cite the datasets and benchmarks you utilize in your report.**

### Primary Evaluation Benchmarks

- **VSI-Bench: A video-based visual-spatial intelligence benchmark containing >5,000 QA pairs derived from egocentric videos. It evaluates 8 tasks across configurational (e.g., Route Plan), measurement estimation (e.g., Absolute Distance), and spatiotemporal categories.**
- **VSTI-Bench: A temporal benchmark designed to test spatial reasoning over time in static 3D scenes based on apparent camera motion.**

**Note that you need to require the datasets by application from the Scannet/Scannet++ which may take a few days, start ASAP.**

### Source 3D Datasets and Additional QA Benchmarks

- **ScanNet: A large RGB-D indoor dataset with 2.5M RGB-D images across 1,513 scans in 707 spaces, with camera poses and reconstructions.**
- **ScanNet++: High-fidelity indoor dataset pairing high-end laser scans with 33-megapixel DSLR images and >3.7M iPhone RGB-D frames.**
- **ARKitScenes: An indoor RGB-D dataset captured using Apple mobile devices with depth sensing and labeled 3D bounding boxes.**
- **ScanQA: A 3D visual question answering dataset based on ScanNet, requiring spatial understanding to answer questions about physical environments.**
- **SQA3D: Situated Question Answering in 3D Scenes, assessing a model's ability to understand 3D scenes from an egocentric perspective.**

- **Video-MME: A comprehensive multimodal evaluation benchmark for video understanding (you may use a subset focusing on spatial/navigation reasoning).**
- **VQA v2: A widely used standard visual question answering dataset useful for baseline comparisons.**

***(Optional Advanced Extension: Navigation episode generation using AI Habitat simulator, generating route-planning trajectories with scene assets from the 3D datasets above.)***

# 3. What You Must Implement

**You will build a lightweight evaluation pipeline to compare models, test specific tasks, and extract failure modes.**

## A. Model Requirements

**Evaluate at least 2 VLMs that support image or video inputs and produce text outputs:**

- **At least 1 open-source VLM (e.g., Qwen3/3.5) running locally (Highly Recommended).**
- **Optionally, include 1 API-based model (e.g., Gemini 3).**

## B. Task Requirements

**Select at least 2 tasks in total. You must pick at least one task from Category 1, and one from Category 2. Report the metrics corresponding to your chosen datasets (refer to the VLM-3R paper for reporting table formats).**

- **Category 1: Navigation & Egocentric Spatial Reasoning (Choose ≥ 1)**
  - **VSI-Bench: Route Plan (multi-step turn/forward planning).**
  - **VSTI-Bench: Camera Movement Direction (relative to start orientation).**
  - **Video-MME (spatial/navigation subset).**
- **Category 2: 3D Scene QA & Metric Geometry (Choose ≥ 1)**
  - **VSI-Bench: Absolute Distance or Room Size.**
  - **VSTI-Bench: Camera Displacement or Camera-Object Absolute Distance.**
  - **ScanQA.**
  - **SQA3D.**
  - **VQA v2.**

## C. Metrics You Must Report

**Compute and report metrics precisely matching benchmark definitions:**

- **Multiple-Choice: Accuracy (exact match to option after deterministic text parsing).**
- **Numeric Answers: Mean Relative Accuracy (MRA), consistent with VSI-Bench definitions.**

- *(Optional)* Additional diagnostics such as confusion matrices or error-vs-distance bin plots.

### D. Failure Analysis Requirements

Extract and analyze explicit failure modes from your inference runs:

- Isolate at least 5 distinct failure cases (include screenshots/thumbnails, question, ground truth, and the model's output).
- Categorize these into 2 to 4 failure types (e.g., scale confusion, egocentric frame confusion, temporal drift, object identity swap).
- For each type, explain *why* it likely occurs and what specific task conditions trigger it.

## 4. One-Page Summary Requirements

Submit a single-page PDF report (compact, technical, heavily utilizing figures and tables. You can have appended pdf as supplementaries, but all of the pages should be within 2 pages). The goal is to communicate results clearly, not to write a long essay.

Your summary must include:

1. Problem Statement (2-3 sentences): Identify the specific spatial weaknesses you targeted and why they matter.
2. Experimental Setup: Detail model names/versions, inference parameters, and the exact benchmark subsets/tasks evaluated.
3. Main Results Table: A concise table mapping Models vs. Tasks, reporting ACC (multiple-choice) and MRA (numeric).
4. Failure Cases Figure: A compact grid (e.g., 2x3 or 2x4) displaying representative visual failures, each accompanied by 1-2 lines of analytical explanation.
5. Short Discussion (5-7 sentences): Synthesize your findings. Distinguish between "easy wins" (e.g., prompt engineering, better parsing) and fundamental model limitations.
6. References: Cite the benchmarks, datasets, and models utilized.

## 5. Grading Rubric (100 Points Total)

- Correctness and Reproducibility (30 pts): Evaluation runs correctly end-to-end. Metric computation strictly matches benchmark definitions (ACC and MRA). You need to find the data source correctly and download, process them for your evaluation.
- Experimental Design (20 pts): Task selection meaningfully tests spatial weaknesses (properly combining navigation and metric/geometry tasks), with strong justification based on benchmark literature.
- Results Quality and Analysis (30 pts): Quantitative results are clear. The failure analysis goes beyond listing errors by defining consistent, evidence-based categories that explain underlying performance gaps.
- Communication Quality (20 pts): The 1-page summary is self-contained, professional, and visually structured (clean tables, clear visual grids, concise scientific writing). Citations are present and

appropriate.

**References (datasets and models you can use)**


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